UNIT QUESTIONS

SKILLS CRITICAL THINKING 4

1 **a** Which of the following is not used to protect us from the possibility of receiving an electric shock?

A double insulation
B live wire
C earth wire
D circuit breaker

(1)

6 **b** Which of the following is true for a negatively charged object?

A It will attract another negatively charged object.
B It has too few electrons.
C It has too many neutrons.
D It has gained extra electrons.

(1)

4 **c** Which of the following is true for all parallel circuits?

A Parts of the circuit can be turned off while other parts remain on.
B The current is the same in all parts of the circuit.
C There is only one path for the current to follow.
D There are no junctions or branches.

(1)

SKILLS PROBLEM SOLVING 7

d When a voltage of 6 V is applied across a resistor there is a current of 0.1 A. The value of the resistor is

A 6 Ω
B 60 Ω
C 16.6 Ω
D 0.6 Ω

(1)

(Total for Question 1 = 4 marks)

SKILLS CRITICAL THINKING 6

2 Copy and complete the following passage about electricity, filling in the spaces.

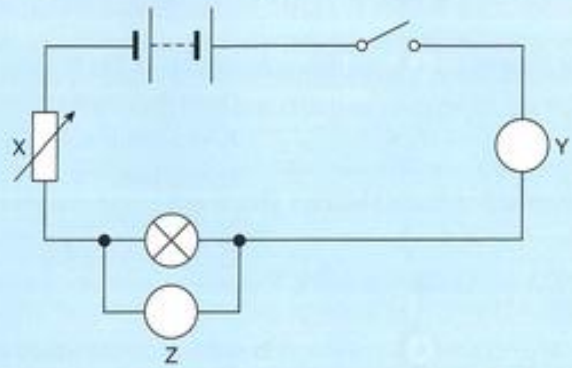
An electric current is a flow of _____. A current of 1 amp is 1 _____ of charge flowing each second. The voltage is the _____ transferred per coulomb of charge.

The current in a component depends on the voltage and the _____; the higher the resistance, the _____ the current.

(Total for Questions 2 = 5 marks)

SKILLS ANALYSIS

3 Asma set up the circuit shown below to investigate how the resistance of a bulb changes as the current in it changes.

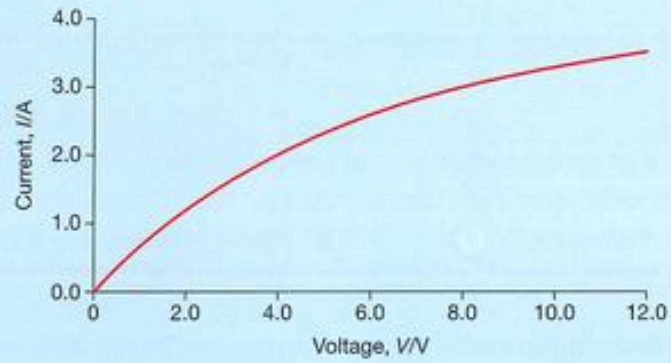


(5) **a** What are the names of the instruments labelled Y and Z? **(2)**

b What is the name of the component labelled X? **(1)**

(6) **c** What is the purpose of X in this circuit? **(1)**

Asma takes a series of readings. She measures the voltage across the bulb and the current in it. She then plots the graph shown below.



d What is the current in the bulb when a voltage of 6 V is applied across it? **(1)**

e What voltage is applied across the bulb when there is a current of 2 A? **(1)**

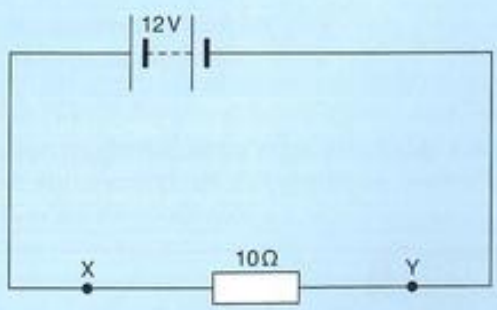
SKILLS PROBLEM SOLVING (7) **f** Calculate the resistance of the bulb when there is a current of 2 A. **(2)**

SKILLS REASONING (5) **g** What happens to the resistance of the bulb as the current increases? **(1)**

(Total for Question 3 = 9 marks)

SKILLS PROBLEM SOLVING

4 A simple series circuit containing a 12 V battery and a 10 Ω resistor was constructed as shown below.



8 **a** Calculate the current between points X and Y. **(2)**

7 **b** Calculate the total charge that flows between X and Y in 5 s. **(2)**

c Calculate the energy transferred in the resistor in 1 minute. **(2)**

(Total for Question 4 = 6 marks)

5 An electric kettle is rated at 2 kW when connected to a 230 V electrical supply.

7 **a** Calculate the current when the kettle is turned on. **(3)**

5 **b** What value fuse should be included in the circuit of the kettle? (Assume that the fuses available are 3 A, 5 A and 13 A.) **(1)**

SKILLS CRITICAL THINKING 6 **c** Modern kettles often have double insulation. Explain what this means and how it provides extra safety for the user. **(2)**

SKILLS PROBLEM SOLVING 8 **d** Calculate the resistance of the heating element of the kettle. **(3)**

(Total for Question 5 = 9 marks)

SKILLS CRITICAL THINKING 6 **6** **a** Explain in detail how insulating materials can be charged by friction. **(4)**

SKILLS REASONING **b** When an aircraft lands it is important that it is earthed before it is refuelled.

7 **i** Explain why the aircraft should be earthed. **(3)**

6 **ii** Suggest one way in which the aircraft could be earthed. **(1)**

7 **c** Explain why electrostatic painting of objects, such as bicycle frames, makes good economic sense. **(3)**

SKILLS CRITICAL THINKING 6 **d** Describe briefly how an inkjet printer makes use of some of the properties of static electricity. **(3)**

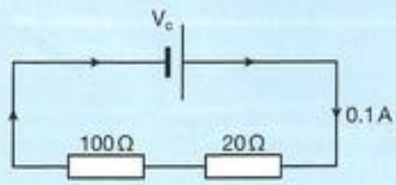
(Total for Question 6 = 14 marks)

5 **7** **a** Describe four uses of the heating effect of electricity in the home. **(4)**

6 **b** Explain why a double-insulated hairdryer does not need an earth wire in its cable. **(2)**

(Total for Question 7 = 6 marks)

SKILLS PROBLEM SOLVING 7 **8** Calculate



a the voltage across the 20 Ω resistor **(2)**

b the voltage across the 100 Ω resistor **(2)**

c the voltage of the cell (V_c). **(1)**

(Total for Question 8 = 5 marks)